R. Brown

Eleventh Annual In-House Convention 37

23 & 24 April 1985

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

TUESDAY, APRIL 23

1:30 - 3:15 Session I. Chair: R. Rodger

1. K. Berridge
2. K.M. Wassenberg, S.E. Bryson & J.F. Werker
3. C.H. Jones
4. R. Brown
5. C. Reed-Elder
6. R. Mistlberger & B. Rusak
7. S.R. Shaw & I.A. Meinertzhagen

COFFEE BREAK

3:30 - 5:00 Session II. Chair: Anastesia Droungas

8. K. Bloom & A. Russell
9. S. Nakajima & I. Faisal
10. K. Briand
11. D. Nicol & I. Meinertzhagen
12. C. Edwards & W. Honig
13. R. Klein

BEER IN LOUNGE

WEDNESDAY, APRIL 24

9:00 - 10:30 Session III. Chair: Ray Klein

14. J. Fentress
15. L. Crowe
16. P. McLeod & R. Brown
17. A. Frohlich
18. J. Fairless & V. LoLordo
19. P.M. Thompson & S.R. Shaw

COFFEE BREAK

10:45 - 12:15	Session IV. Chair: John Barresi	
		20. J. Radel 21. D.B. Henken 22. J. McNulty 23. S.G. Marlin & V.M. LoLordo 24. B. Rusak 25. W.K. Honig
12:15 - 1:30	LUNCH	
1:45 - 3:30	Session V. Chair: Aaron Gilani	
		26. J. McLaren 27. D.E. Mitchell 28. J. Connolly 29. M.J. MacLean & S.E. Bryson 30. C.L. Baker, Jr. 31. R. Croll
3:30 - 3:45	COFFEE BREAK	
3:45 - 5:15	Session VI. Chair: B. Rusak	
		32. M. Spetch & C. Edwards 33. J. Enns 34. D.P. Phillips 35. A.R. Delamater, V.M. LoLordo K.C. Berridge 36. K. Murphy 37. S. McKelvie
5:30	BEER & DINNER	

ABSTRACTS

1. K. Berridge

Modulation of Palatability by Trigeminal Tactile Sensation

The trigeminal (5th cranial nerve) system carries somatosensation from the face and mouth. Trigeminal sensation is crucial to normal feeding. This experiment examined in rats how trigeminal tactile input modulates the perceived palatability of tastes, as measured by species-typical actions that are sensitive to palatability evaluations. The results suggested that low-level trigeminal input amplifies selectively positive palatability while having little or no effect on the aversive limb of palatability evaluations. Trigeminal deafferentation thus unbalances the palatability of tastes.

2. K.M. Wassenberg, S.E. Bryson & J.F. Werker

Consonant Substitution Errors in Reading-Disabled and Normal Children: Phonetic, Visual and Frequency Effects

Consonant misreadings were examined in two groups of severely reading-disabled children, and both age- and reading-watched, normal controls. Stimuli were single-syllable nonsense words with consonants in word-initial and/or -final position.

Substitution errors were classified according to: i) word-position; ii) phonetic simularity; ii) visual similarity. Comparisons between groups showed effects of position and phonetic similarity for the normal children. The reading-disabled groups showed no effect of position, a strong visual effect and differing patterns of phonetic error types.

In addition, in both reading-disabled groups, unlike the normal children, /d/ was the most frequent error response. These results are discussed in terms of the articulatory hypothesis (Werker & Bryson, 1984).

3. C.H. Jones

The Effect of a Sandy Substrate on Intra-Specific Agonistic Behavior in the Syrian Golden Hamster (Mesocrioetus auratus).

The agonistic behavior of male Syrian golden hamsters was recorded in environments with sandy and barren substrates. The effect of the sandy substrate was to decrease the amount of agonistic behavior shown by the hamsters in comparison to levels observed when the substrate was barren. This effect was observed across varying motivational states, prior housing conditions of the animals, lighting conditions of the test arena, light dark cycles and social groupings during the test. The results are discussed in terms of traditional conceptualizations of the hamster's social tendencies in the wild.

4. R. Brown

Dissecting the Components of the Maternal Odour of Mice

In a follow-up to my last in-house presentation, we have conducted another five experiments on the attraction of 16-20 day old mice to "maternal odours" and I shall discuss the results of these with reference to last year's results. Using feces from mice as stimuli in a series of forced choice tests, we have examined the effects of diet (Purina vs. Halifax Aquarium House), species (rat vs. mouse), sex (male vs. female), lactation (lactating vs. nonlactating females) and relatedness (own vs. another mother) on odor preferences. Each of these factors appears to contribute to the attractiveness of the maternal odour, with diet playing an especially important role. These results indicate that the "maternal odour" is a complex stimulus which includes many components and that pups may attend to some of these components more than others. Cheryl Reed-Elder will give an interpretation of these results.

5. C. Reed-Elder

Is the Concept of a Maternal Pheromone useful to explain the Attractiveness of Maternal Odours?

Studies of odour preferences were examined in an effort to answer some questions regarding the interpretation of such preferences as evidence for a "maternal pheromone". Some odorous substances, in particular, feces and ventral scent gland sebum, both of which are copiously produced by lactation females, are preferentially approached by preweanling pups. The attractiveness of an odour to pups may, in part, be a function of the odour's saliency, its familiarity, or its similarity to other, already familiar odours. Unfamiliar odours can be made attractive by exposure, and familiarity with an odour seems to be both necessary and sufficient to increase the attractiveness of that odour, whether it is produced by a conspecific or not. Simple exposure has been shown to be effective in establishing an odour as attractive but this attraction may be increased if conditioning contingencies are also present. Since the attractiveness of maternal odours can be explained simply by assessing the role of familiarity, the concept of a "maternal pheromone" is not necessary and is not parsimonious with the data. Furthermore, it is not successful in explaining the modifiability of odour preferences through experience.

6. R. Mistlberger & B. Rusak

Effects of Paraventricular Nuclei Lesions on Entrainment of Circadian Activity Rhythms to Restricted Feeding in the Rat

Periodic food availability cycles (2h food - 22h fast) can entrain a variety of circadian rhythms in rats. The phase angle of entrainment is such that these rhythms anticipate the daily mealtime; for example, the rat becomes active 2-3h before mealtime. The neural mechanisms mediating anticipatory, food entrainable rhythms are functionally and antomically distinct from those controlling free running, light entrainable rhythms since suprachiasmatic nuclei lesions abolish or severely disrupt free running but not food anticipatory rhythms. Previous attempts to localize critical components of the food entrainment mechanism indicate that the pineal and adrenal glands, the hypophysis and the ventromedial hypothalamus are non-essential. The present study investigates whether the paraventricular nuclei (PVN) might be involved, in view of their purported role in the regulation of food intake and the preminence of food intake over light in the phase control of certain PVN receptor rhythms. Circadian locomotor rhythms were monitored in 9 rats on restricted food schedules (2h diurnal mealtime) for 2 weeks prior to and 3 weeks after bilateral PVN ablation. Six rats displayed signs of obesity typically associated with PVN lesions. Anticipatory activity was markedly attenuated in 3 of these rats and largely unaffected in the remaining 6. The animals will be tested again after several months to assess the stability of this effect.

7. S.R. Shaw & I.A. Meinertzhagen

Synaptic Promiscuity and Neuronal Fidelity as the Forces of Destiny.

Comparative ethologists have long suggested that animal behaviour has evolved progressively as speciation has proceeded. Changes in behaviour may have been just as important in evolution as the more obvious alterations in body plan, for instance as mechanisms ensuring reproductive isolation. By contrast, neuroscientists have had little that is useful to offer about the evolution of the brain, which after all supposedly controls behaviour. They offer unlikely tales that changes in neuron size and shape, in brain size or in the direction taken by neuronal tracts, are the main correlates of brain evolution. A more appealing idea is that new neuronal classes arise at each twist and turn of evolutionary advance, accounting for each new behaviour, but this appears confounded by the facts. In cases where we can identify neurons as individuals, they appear to have persisted as recognizable entitles for perhaps >100 million years (admittedly few cases so far, mostly in invertebrates). How might this apparent bottleneck to the evolution of brain function be overcome? If "new" neurons cannot readily emerge in the CNS, we speculate instead that rearrangement of synaptic connections amongst existing nerve cells must take precedence, as a fundamental mode of evolutionary progression. Evidence supporting this new if obvious hypothesis will be presented, drawn from a comparative EM study of a particular class of sensory synapse that is made between identifiable neurons, and that must have existed since the Carboniferous (~250MYr BP). We have discovered that the composition of the post-synaptic set of elements at the synaptic has altered, during the monophyletic evolutionary divergence into the many modern families. Alterations in some other types of synapse have also been noted.

8. K. Bloom & A. Russell

Longitudinal Study of Social Vocalizations of Infants

Recently we demonstrated that 3-mos-old infants produce vowel-like (vocalic) sounds when adults maintain the conversational rules of turn-taking inherent in social contingencies. The present experiment demonstrated the developmental course of this effect by randomly assigning 7-wk-old infants to one of 3 groups. In 6 weekly sessions infants received adult stimulation contingently, yoked, or randomly related to their vocalizations. Across the 6-wk period all infants equivalently increased their overall vocal rate, but the proportion of vocalic to nonvocalic sounds was significantly highest for the contingent turn-taking on the expressive quality of infant vocalizations is present as early as the 7th wk of life and reflects an adaptive process of early social development.

9. S. Nakajima & I. Faisal

Dopamine Receptors Involved in Food Reinforcement

Rats trained to press a bar for food pellets under CRF, VI-1 min, or VI-3 min schedule were injected with SCH 23390 (dopamine D1 receptor blocking agent), sulpiride (D2 blocker), or physiological saline. D1 blocking suppressed responding, but D2 blocking did not. Responding under the VI schedules was more susceptible to D1 blocking than the responding under CRF.

10. K. Briand

Attention and Feature Integration

Treisman's Feature Integration theory of attention predicts that focal attention is needed to correctly combine the different features which make up a visual pattern or object. The present study was intended to determine whether orienting of attention to (or away from) a given spatial location would affect the tendency for subjects to report illusory conjunctions at that location. Subjects searched for the letter R with two sets of distractors, PB and PQ. Results showed that performance was better when the location cue was valid than when it was invalid, and importantly, that this effect was no greater for the PQ distractor set than for the PB distractor set. The data are inconsistent with the notion that allocating attention spatially serves a role in integrating the features of an object correctly.

11. D. Nicol & I. Meinertzhagen

The Cell Lineage of the Ascidian Neural Plate

Ascidians, or sea-squists, are unprepossessing sessile, marine animals which produce embryos demonstrating almost perfect mosaic development. The embryo produces a motile larva with chordate-like characteristics including a dorsal tubular nerve cord which forms from a neural plate and on the basis of which the group is thought to resemble the primitive chordate ancestor. We have examined the lineage of the cells of this neural plate and have now accounted for all the cell divisions (about 12) which generate all the cells (about 350) of the nervous system. We are thus how ready to examine the effects of cell deletions and thus test the ability of the neural plate to regulate both the cell number and cell type of its progeny.

12. C. Edwards & W. Honig

Category Learning in Pigeons

Four groups of pigeons were trained on a go/no go task that required birds to discriminate slides of naturalistic stimuli. Pairs of slides had identical backgrounds, but within each pair the slides were distinguised by the presence or absence of a human figure. The group of birds trained to respond to human present slides acquired the discrimination. However, a second group, one that was trained to respond to human absent, failed to acquire the discrimination. Two additional groups that were given "pseudo-concept" training (i.e., both the positive and the negative slide sets contained human-present and human-absent slides) also failed to acquire the discrimination.

13. R. Klein

<u>Stimulating the Subjective Lexicon</u>

Galton's studies of associative processes suggested a complex mental machinery that seemed unavailable to conscious introspection. That machinery which Galton likened to the "pipes and flues and bell wires" in one's basement might today be called the subjective lexicon (G. Miller) or semantic memory (Quillian, Tulving). The most popular paradigm for exploring properties of the subjective lexicon has been to examine the effects of prior or simulataneous stimulation of some region of the lexicon upon task performance involving the same or different region. The evolution and current status of this paradigm will be described and exemplified with some recent data from our lab.

14. J. Fentress

<u>Modulations Muddle Monolithic Modules in Mammamilian Movement</u>

As in figurative tongue twisters, movements of the tongue and other appendages in rats show incomplete encapsulation (modularity). Kent Berridge and I have recently been able to show this by examining the detailed <u>form</u> of facial and forelimb actions in trigeminally deafferented animals under various motivational and sequential contexts. The point of the exercise is not merely that strict dichotomies between "reflex" and "centrally patterned" actions are both a logical and biological misnomer, but that even relatively simple behavioural patterns exhibit dynamic and relational control preperties that deserve further investigation. Contextual factors are necessary complements to more traditional (fragmented) conceptualizations of control.

15. L. Crowe

<u>The Effect of Viewing Pornographic, Violent and Pornographic Violent Film on Viewer's Subsequent Administration of Reward and Punishment</u>

Twenty-seven undergraduate males participated in this study designed to examine the effect of pornographic and pornographic violent film on viewer's tendency to reward and punish a female. Following exposure to either a pornographic, pornographic violent, neutral or violent 10 minute videotape segment, subjects had the opportunity to reward and punish a mythical female learner in a bogus teaching task. The effect of the film was measured as the ratio of the total duration of rewards to the total duration of punishment. Sequential analysis ($\alpha = .05$) indicated significant differences in the reward to punishment ratios for all film conditions. The highest ratio was found for the pornographic film and the lowest for the pornographic violent film. Results are discussed in terms of the aggressive cue value of the films. Several explanations, including reference to the social learning theory, were offered to account for this unexpected pattern of results.

16. P. McLeod & R. Brown

<u>Effects of Prenatal Stress on Parental Behavior of Male Rats</u>

Prenatal environmental stress has been shown to both feminize and demosculinize the sexual behavior of male rats (Ward, 1972). The extent of these effects is influenced by prepuheral rearing conditions (Ward and Reed, in press). We are conducting an experiment on another sexually dimorphic behavior. This tests the interaction between prenatal stress and post-weaning social environment or parental behavior of male rats. Data available from this experiment (which is currently in progress), will be presented.

17. A. Frohlich

<u>Developmental Perturbations in Postsynaptic Cells Produced by Cold-Shock during Development</u>

After studying normal development leading to the formation of tetrad synapses by the visual receptors in the complex eyes of the house fly, we have now started to disturb the developmental processes to further unravel developmental rules governing synapse formation. In a pilot study using (for the time being) the sledge hammer method of applying heat- or cold-shocks during development we hope to disturb the formation of the tetrad synapse, for instance by knocking out propective post-synaptic cells. Preliminary results are presented.

18. J. Fairless & V. LoLordo

<u>The Effect of Context Shifts upon the Resistance-to-Reinforcement Assay of Conditioned Inhibition</u>

Pigeon's exposed to a negative CS-UCS contingency show attenuated acquisition of excitatory responding when these stimuli occur later in a positively contingent relation. This effect constitutes the traditional resistance-to-reinforcement assay of conditioned inhibition. Our most recent work suggests that his acquisition deficit may be composed of three separable processes, only one of which is attributable to pre-exposure to the negative CS-UCS relation per se.

19. P.M. Thompson & S.R. Shaw

<u>The Holly Grail: A Quest for Circadian Rhythmicity in the Life of a Fly</u>

Nassel et al recently discovered an efferent serotonergic neuron TAN-3, that runs out from the brain to end in diffuse branches in the fly's peripheral visual system. The structure and location of TAN-3's endings suggest that it releases its neurotransmitter non-synaptically throughout the eye, perhaps such as to influence some "global" attribute of visual processing. Since very large circadian changes in visual sensitivity are found in the eyes of many arthropods at dawn and dusk, and are mediated by bioamine-containing efferent neurones in one known case, we hypothesized that TAN-3 could be another such circadian controller neuron. We shall describe long-term retinal recordings made to test this idea, that ought to reveal any circadian changes in visual sensitivity taking place either in the photoreceptors themselves, or in the neurons immediately post-synaptic to them. This being a psychology department, we also looked to see if this strain of flies showed any behavioural evidence for circadian rhythmicity. Actually, our most intriguing finding has nothing much to do with all this: it is that long term light-adaptational depression follows exposure to simple room lights. Adaptational recovery is quite fast at the level of the photoreceptors, and the long term changes therefore must have been added at the synaptic zone itself. This suggests, tentatively, a different "global" role for a neuron like TAN-3 in visual processing, that should be testable in future.

20. J. Radel

Regeneration in the Central Nervous System of Small, Furry Creatures

Traditional wisdom states that the mammalian central nervous system is incapable of regenerating damaged tissues. I will present a brief history of the more successful attempts to induce CNS regeneration in mammals.

21. D.B. Henken

An Examination of Cell Proliferation in the Goldfish Retina During Regeneration of the Optic Nerve

Using thymidine autoradiography, I am examining changes in the rate of cell birth in the retina as a consequence of optic nerve crush. The question that I am addressing is, does regeneration of retinal ganglion cell axons influence other cells that are one or two connections removed from the event? I have found changes in cell birth rate in the outer nuclear layer and ganglion cell/optic fiber layer of the retina that appear to coincide with the regeneration process.

22. J. McNulty

Experiments in the use of Hyperbaric Oxygen for the Treatment of Medical Conditions

Hyperbaric oxygen is used as the primary mode of treatment for conditions such as decompression illness and as adjunct therapy for a wide range of medical conditions from asteromyelitis to skin grafts. There is also a class of conditions categorized as experimental, for which these are suggestions that hyperbaric oxygen may be of benefit but for which further research is required. Research currently being done at the Hyperbaric Medicine Unit of the VG Hospital for one of these "experimental" conditions will be described.

23. S.G. Marlin & V.M. LoLordo

Endogenous Pain Control Systems: Excitation and Inhibition Procedures for Classically Conditioned Analgesia

By repeatedly pairing a neutral cue with a noxious stimulus, the cue alone comes to evoke analgesia. To examine conditioned excitation of analgesia rats were restrained and received tailshock during the last 15 seconds of a 1.00 minute white noise CS presentation. Following three and six trails, the animals pain reactivity during the CS was assessed using the tail-flick test. The rats showed increased pain thresholds relative to baselines after six but not after three trials. A second study examining the time course of the decay of analgesia following four conditioning trials showed profound analgesia during the CS and for 14 min following the test trial. Also the possibility of producing conditioned inhibition using this paradigm was investigated. For a stimulus to be a conditioned inhibitor it must counteract a conditioned excitor. To achieve this a flashing light/white noise compound stimulus was added which reliably predicted the absence of shock. Experimental animals showed higher pain thresholds to the excitor than to the flashing light/white noise compound. The consistancy of these results suggest conditioned inhibitor mechanisms were responsible. Results are discussed in terms of Bolles and Fanselow's perceptal-defense-recuperative model of fear and pain.

24. B. Rusak

Rhythms, snake oil and acetylcholine

The neurotransmitter acetylcholine has been implicated in a number of relatively trivial functions, such as causing muscular contractions and maintaining higher mental functions in humans. It has now been implicated in entrainment of rodent circadian rhythms, and has therefore finally attracted my attention. I'll review some older data suggesting a cholinergic mechanism for light entrainment of rhythms, some newer data indicating some strange complexities in putative cholinergic binding sites in the CNS, and I'll talk about some ongoing and planned studies related to these issues.

25. W.K. Honig

Modulation of Performance in Working Memory as a Function of Anticipated Frequency of Long and Short Memory Intervals

Pigeons are learning two delayed discriminations concurrently; blue and white are the initial stimuli in one; green and red in the other. The same long and short memory intervals (MIs) are used in each problem (e.g., 2 sec and 10 sec). However, one problem contains 80% long MIs and 20% short MIs, while the other contains the reverse proportions. We expect that performance at a given MI will be modulated by the anticipated frequency of that interval; pigeons should do worse with the long MI if it occurs on 80% of the trials in one problem than if it occurs on 20% of the trials in the other.

26. J. McLaren

Hemispheric Differences in the Perception of Happy and Sad Faces: Effects of Expressive Asymmetries in the Stimuli

The purpose of the present series of experiments was to further evaluate the claim that the two cerebral hemispheres are differentially specialized for the perception of positive and negative emotion (cf. Reuter-Lorenz and Davidson, 1981). In a previously reported study we failed to replicate the finding of a left visual field advantage for sad faces and a right visual field superiority for happy faces. Our study differed from that of others in some methodological respects. Most importantly, we used mirror-image as well as normally orientated faces, to control for expressive asymmetries in the stimuli. The experiments to be reported were designed, firstly, to investigate the apparent discrepancy described above and, secondly, to determine the effect of subject's mood on the lateralized perception of emotional faces.

27. D.E. Mitchell

A Cure for Amblyopia

It is well known that a period of monocular deprivation in early life can exert profound effects on the development of the visual pathways of kittens. These are accompanied by severe deficits in the vision of the deprived eye that are comparable to those observed in severe forms of human amblyopia. Although these effects are severe, considerable recovery of vision can occur in the deprived eye if, subsequent to the period of monocular deprivation, the formerly nondeprived eye is occluded. However, we have reported here earlier that the recovery promoted by this procedure (reverse occlusion) is not maintained afterwards. We now report that it is possible to achieve total behavioural recovery by instituting part-time occlusion where the nondeprived eye is occluded for only part of each day followed by a daily period of at least two hours during which both eyes are open. Thus while only limited permanent recovery is achieved in situations when either of these conditions are imposed full-time, a combination of both can lead to complete behavioural recovery for certain conditons of deprivation.

28. J. Connolly

Psychophysioalogical Research in Schizophrenia

A brief overview of the rationale and objectives of psychophysiological research in the schizophrenias will be presented. An example of basic psychophysiological research applied to psychopathology will be discussed. Recently completed work on event-related brain potentials relevant to the dichotic listening procedure together with in-progress tachistoscopic data will be presented. These and other data will be used to suggest that so-called "higher order cognitive deficits" seen in schizophenia are attributable to "lower order" abnormalities in sensory and perceptual processes.

29. M.J. MacLean & S.E. Bryson

Differing Emotional Responses of the Two Cerebral Hemispheres: The Effects of Scale and Gender

Previous authors (see, e.g., Natale, Gur & Gur, 1983) report that the two cerebral hemispheres differ in their emotional responses. The general finding is that stimuli projected to the right field-left hemisphere are rated more positively than those projected to the left-field-right hemisphere. Most studies have employed left-to-right positively ascending scales. The present series of experiments was carried out to evaluate the effects of scale on judgements of emotion. Sad, happy and neutral faces were presented unilaterally and centrally and right-handed subjects rated the degree of happy or sad emotion on one of four types of scales. Our results qualify previous findings in this area. They suggest that careful attention must be paid to the effects of spatial compatibility in laterality research.

30. C.L. Baker, Jr.

Spatial properties of direction-selective neurons in striate cortex of the cat

Bar-shaped stimuli flashed sequentially at adjacent points in the receptive fields of direction-selective striate cortex neurons produced direction-selective behavior comparable to that observed with smoothly moving stimuli. Independent variation of the positions of the 2 flashes allowed determination of the optimal displacement, D_{opt}, for direction-selectivity from sequential flashes. Measured values of D_{opt} typically ranged from 0.3 to 0.8 deg for receptive field sizes of ca 4-7 deg.

The spatial separation of 2 simultaneously presented bars which produces the largest lateral inhibition was also determined for each direction-selective neuron. Twice this value provided the spatial subunit wavelength, λ, for a given neuron. Measurements of λ were obtained from line weighting functions, in the case of Simple cells, or from double-line interaction functions, in the case of Complex cells[1]. Measured values of λ were generally ca 1-5 deg.

For the majority of neurons studied, the ratio of optimal displacement for direction-selectivity (D_{opt}) to subunit wavelength (λ) fell in the range of 0.10-0.35. These results imply a systematic quantitative relationship between the spatial subunit wavelength and mechanisms for direction-selectivity.

Supported by Centennial Fellowship (Canadian MRC) to CLB and a Canadian MRC (PG-29) to MC.

31. R. Croll

Functional anatomy of gastropod neurons

While gastropods have been used extensively for the last two decades as preparations for studying the physiological bases of behaviour, relatively few techniques are available for examining the details of neuronal morphology in this class of animals. Over the last year I have been applying two techniques to the gastropod nervous system which reveal much new information about the functional anatomy of single identifiable neurons.

32. M. Spetch & C. Edwards

The "Milk Run": A New Apparatus for Studying Spatial Memory in Pigeons

Pigeons, unlike rodents, do not display good spatial memory in traditional radial-arm mazes without special and extremely lengthy training. Thus, they seem able, but "unprepared" to use spatial memory in radial mazes, perhaps because labyrinthine structures are constraining for pigeons who, in the wild, generally feed in open fields. In support of this, pigeons displayed spatial memory without extensive pretraining in an open-room "flight maze" (reported at the 1984 In House). However, since some birds proved reluctant to make the flight response, we have recently developed an open-field, ground feeding version of the "maze" that involves a number of "feeding stations" (milk cartons baited with peas) placed on the floor. Our preliminary results suggest that this apparatus overcomes the response problems of the flight maze, and that pigeons readily display good spatial memory to efficiently locate the food.

33. J. Enns

Further Explorations of Changes with Age in the Integration of Shape

Experiments with observers aged 6-24 years examined changes with age in the sequential integration of shape in a free-viewing task. The stimuli were line-drawings of "ribbons" which consisted of a series of panels connected randomly at 90° angles. The task was to indicate whether the end-panels of each ribbon displayed the same surface or different surfaces to the viewer. Age-related improvements were found in the speed and accuracy of this task (Experiment 1). Moreover, these improvements with age were found even when the distance over which observer's eyes travelled was held constant (Experiment 2) and when the need for any overt eye movements was eliminated (Experiment 3), supporting the view that the ability to construct cognitive "schemas" of novel objects improves with age. A final experiment explored the effects of instructing observers to use special strategies to solve the task.

34. D.P. Phillips

Forward Masking Studies of Cat Auditory Cortex Cells

Single neurons in the cat's auditory cortex encode stimulus intensities over limited ranges. The "dynamic range" undergoes a shift towards higher sound pressure levels if the stimulus of interest is presented in a background of continuous white noise. The magnitude of the dynamic range shift is linearly related to the intensity of the noise, and is comparable in cells that are excited or inhibited by noise bursts. To examine the sequence of neural events underlying this dynamic range shift, cortical cells were studied parametrically with preferred-frequency tones delivered at various delays after the onset and offset of a noise mask. These experiments shed light on the mechanisms used by the central nervous system to defect signals in noise.

35. A.R. Delamater, V.M. LoLordo & K.C. Berridge

Exteroceptive Pavlovian Signals Can Control Palatability

Whereas the associative modulation of "palatability", i.e.,taste hedonics, has been considered to depend upon the relation of only particular types of stimuli (e.g., tastes and illness), the present experiments ask whether exteroceptive conditioned stimuli (e.g., tones) can enter into association with tastes and come to alter the palatability of water. In the first experiment, when a tone predictive of sucrose preceded the oral infusion of water, the tone elicited more sucrose-typical consummatory responses than if no signal preceded or if a previously nonreinforced signal preceded water infusion. In the second experiment, when conditioned stimuli predictive of either sucrose or quinine preceded water infusion, the responses to water were sucrose-typical or quinine-typical, respectively. These data suggest that the Pavlovian mechanisms involved in altering the palatability of tastes are not evolutionarily constrained so as to exclude the potential importance of exteroceptive conditioned stimuli. In addition, they point to a possible role for supporting stimuli in the manifestation of associations.

36. K. Murphy

Ocular Dominance Changes Following Reverse Occlusion in the Kitten

Abnormal visual experience during the critical period has a profound effect upon the functional development of the visual system. Monocular deprivation shifts the ocular dominance and visual acuity in favor of the nondeprived eye. However, these shifts may be reversed if vision is restored to the deprived eye sufficiently early and at the same time the other eye is occluded (reverse occlusion). When a kitten is monocularly deprived until 5 weeks of age, only 18 days of reverse occlusion is necessary for a total ocular dominance shift in favor the initially deprived eye. If binocular visual experience is given subsequent to this length of reverse occlusion both eyes recover only rudimentary visual acuity (1-3 cycles/deg.). This is a very puzzling finding since binocular vision is initiated well within the critical period. To further investigate this phenomenon single unit recording from area 17 was made either immediately upon termination of reverse occlusion or following one month of binocular experience. After binocular experience the distribution of ocular dominance is virtually normal, which leaves us without a physiological correlate for the poor visual acuity. These findings hold important implications for the nature of the mechanisms underlying the physiological changes that occur during reverse occlusion and the methods by which cortical function should be assayed. Also, preliminary results of the arborization pattern of the geniculo-cortical afferents following this deprivation regimen will be presented.

37. S. McKelvie

Trace Acquisition with and without Awareness

One hundred and sixty-three subjects were tested individually on the Hebb digits task in which they listened to 24 lists of 9 digits read at a rate of about 1 per sec and following each list, attempted to recall the digits in their correct order; however, they were not informed that the series on Trial 3 was repeated on every subsequent third trial. On the basis of a post-experimental questionnaire, subjects were categorized as Unaware (*n* = 21), Late Aware (*n* = 29), and Early Aware (*n* = 29) with respect to the reoccurrence of the repeating string. Similar rates of learning of the critical series were found for the three groups, supporting the contention that repetition of a set of digits casn directly set up a structural or long-term memory trace.